Online Retail Business Analytics Report

Revenue, sales, customers, products and market intelligence · Analytical report based on the supplied OnlineRetail_Cleaned.csv and online retail re.sql

1. Executive Summary

Between 1 Dec 2010 and 9 Dec 2011 the dataset records **£10,247,219** of gross product sales, **£475,901** of recorded cancellations (4.6% of gross) and **£9,771,318** of net product revenue across **19,773** sales invoices and **4,362** identified customers. Net revenue is net of recorded cancellations only and is *not* verified accounting net sales. Currency is not stated in the file; GBP is assumed.

Key findings

• **Growth is concentrated in autumn.** Net revenue rose from £701,290 (Aug 2011) to £1,427,125 (Nov 2011), with Nov up 34.4% on Oct. With only one year of data this is consistent with a pre-holiday build but cannot prove seasonality.
• **The business depends on the UK.** The UK provides 84.8% of net revenue; the next four markets (Netherlands, EIRE, Germany, France) each contribute 1.9-2.9%.
• **Revenue is broad-based across products, concentrated in customers.** The top 10 of 3,921 stock codes earn 8.2% of product revenue, but the top 20% of codes earn 78.4%. The top 10 customers hold 16.6% and the top 1% hold 30.2% of identified spend.
• **Retention is a lever.** 65.3% of customers placed more than one sales order. 1,449 customers (about £871,183 observed spend) had not purchased in the 90 days to the last data date.
• **Two large invoices distort gross sales.** Invoices of £77,184 (Jan 2011) and £168,470 (Dec 2011) were fully cancelled by matching C-invoices; gross sales overstate those months and returns look higher (4.6%) than they otherwise would (about 2.3% excluding these pairs).
• **The SQL discount proxy is not reliable.** Using the maximum observed price as list price implies a discount larger than half of notional sales. It should not be reported as discounting.
Decisions for management: prioritise win-back of high-value inactive customers, reduce single-market dependence deliberately, investigate cancellation drivers, and obtain real price and cost data before reporting discount or profit KPIs.

2. Dataset Overview

Item	Value
Records	534,129
Distinct invoices	23,796 (incl. 9,251 cancellation lines on C-invoices)
Customers / products / countries	4,371 / 3,938 / 38
Period	2010-12-01 08:26 to 2011-12-09 12:50 (Dec 2011 covers only 9 days)
Lines without CustomerID	132,565 (24.8%; 15.4% of product revenue)
Service / fee lines	2,924, net £-23,187
Checks passed	0 duplicate rows; 0 zero/negative prices; 0 revenue mismatches (Revenue = Qty × Price); every negative quantity sits on a C-invoice

Data-quality treatment (no records were deleted or altered)

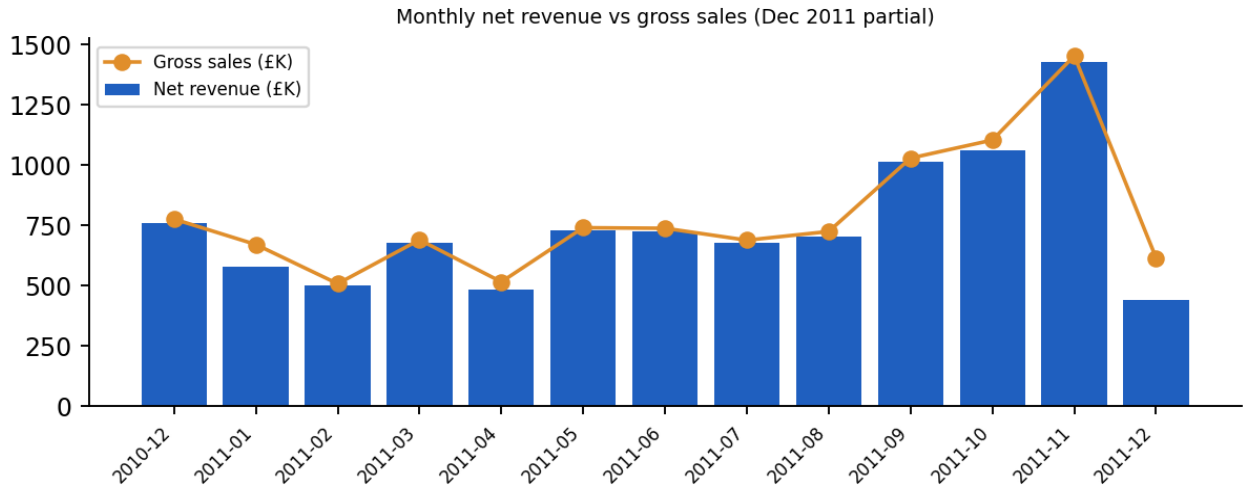
Issue	Impact	Treatment
Cancellation invoices (C…)	Reduce revenue; cannot be linked to originals; returns vs cancellations indistinguishable	Reported separately as Returns; gross, returns and net all shown
Service/fee codes (POST, DOT, M, AMAZONFEE, BANK CHARGES, CRUK, C2, D, S, B, PADS, gift_*)	Not product sales; AMAZONFEE alone is -£221.5K, DOT +£206.2K	Excluded from product KPIs; reconciled to SQL total
Missing CustomerID	Customer metrics cover 84.6% of product revenue	Excluded from customer KPIs only

Issue	Impact	Treatment
Multiple descriptions per stock code (217)	Splits products when grouped by description	Group by StockCode, most frequent description
Partial last month	Understates Dec 2011 (9 days)	Flagged in dashboard; excluded from decline lists
Extreme transactions (±£77K, ±£168K)	Inflate gross sales in two months	Retained; shown on Risks page

3. KPI Performance

KPI	Value	Definition
Net revenue	£9,771,318	Product sales less recorded cancellations
Gross sales	£10,247,219	Product sales lines, C-invoices excluded
Returns value	£475,901	Absolute C-invoice lines
Orders	19,773	Distinct invoices with product sales
Purchasing customers	4,362	Distinct non-null CustomerID
Units sold (gross / net)	5,561,384 / 5,291,367	Sum of Quantity
Average order value	£518.24	Gross sales / sales invoices
Avg revenue per customer	£1,894.88	Identified net revenue / identified customers
Repeat purchase rate	65.3%	2,829 of 4,334 customers with >1 sales order
Avg selling price	£1.84	Gross sales / gross units
Est. gross profit (assumption)	£3,908,527	Net revenue × 40% (from SQL script; not actual profit)
Revenue growth	N/A	No earlier period exists for the full range; period comparisons are available in the dashboard

4. Revenue and Sales Trends



Monthly net revenue was £578,855 in Jan 2011, dipped in Feb (£499,464) and Apr (£482,062), held between roughly £680K and £730K from May to Aug, then climbed to £1,011,340 (Sep), £1,061,701 (Oct) and £1,427,125 (Nov). Orders followed the same shape (peak 2,751 in Nov), so growth is volume-driven rather than price-driven: monthly AOV moved between about £417 and £620 with no clear trend while orders rose from 1,339 in Aug to 2,751 in Nov. Dec 2011 covers only nine days and must not be read as a decline. Seasonality cannot be established from a single year.

5. Product Performance

Top by revenue	Net rev.	Top by units	Units
Regency Cakestand 3 Tier (22423)	£164,459	Popcorn Holder (22197)	56,427

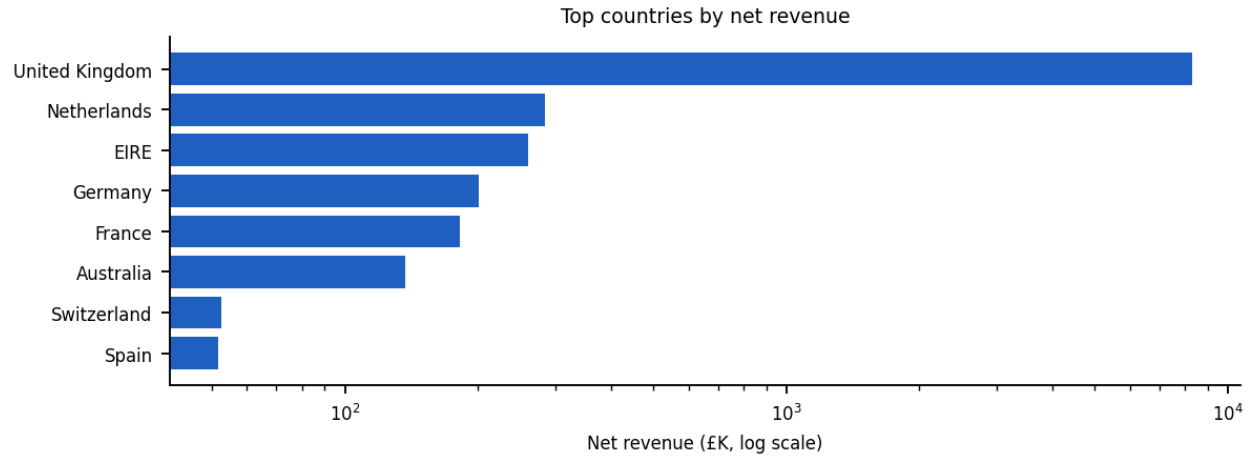
Top by revenue	Net rev.	Top by units	Units
Party Bunting (47566)	£98,244	World War 2 Gliders Asstd Designs (84077)	53,751
White Hanging Heart T-Light Holder (85123A)	£97,838	Jumbo Bag Red Retrospot (85099B)	47,256
Jumbo Bag Red Retrospot (85099B)	£92,176	Assorted Colour Bird Ornament (84879)	36,282
Rabbit Night Light (23084)	£66,662	Pack Of 72 Retrospot Cake Cases (21212)	36,016
Paper Chain Kit 50'S Christmas (22086)	£63,715	White Hanging Heart T-Light Holder (85123A)	35,063
Assorted Colour Bird Ornament (84879)	£58,792	Rabbit Night Light (23084)	30,631
Chilli Lights (79321)	£53,747	Mini Paint Set Vintage (22492)	26,437

Rankings differ because price differs: Popcorn Holder leads on units (56,427) but is tenth on revenue, while Regency Cakestand 3 Tier leads revenue with far fewer units. Revenue is broad: the top 10 codes are only 8.2% of product revenue, though the top 20% earn 78.4% (Pareto). Low revenue does not imply low profit; no cost data is available. Estimated product gross profit uses the disclosed 40% assumption.

6. Customer Insights

4,362 identified customers generated £8,265,477 (observed spend within 13 months, not predictive lifetime value); average revenue per customer is £1,895. 65.3% are repeat buyers. The largest observed spenders are customers 14646 (£278,778, Netherlands), 18102 (£259,657) and 17450 (£189,576); customer 14911 (EIRE) placed 198 orders. Using the last data date as observation date, 1,449 customers are inactive for more than 90 days with £871,183 observed spend. A 90-day threshold over a 13-month window includes many one-time buyers, so prioritise by prior spend and frequency: the largest inactive customers are 15098 (£39,620, 182 days), 15749 (£21,536, 235 days) and 16180 (£10,217, 100 days).

7. Geographic Performance



The UK dominates (84.8%). Some smaller markets show very high AOV, notably Netherlands (£3,053 across 93 orders) and Australia (£2,466 across 56 orders), but they rest on only 9 customers each; EIRE has 282 orders from 3 customers. These are concentrated accounts, not broad markets, so country AOV or revenue-per-customer comparisons should be read with the sample size in view. Germany (443 orders, 95 customers) and France (382, 87) are the broadest non-UK markets.

8. Profitability and Discount Analysis

No cost, margin or list-price data was supplied. Estimated gross profit of £3,908,527 is simply net revenue × 40%, carried over from the SQL script as a disclosed assumption. The SQL discount query takes MAX(UnitPrice) per stock code as list price; on this data it implies about £16,044,136 of notional discount (61% of notional sales), which is implausible and driven by isolated high prices. It is retained only for reconciliation and should not be used to claim discounting behaviour.

9. Risks and Data Quality

Cancellations cannot be matched to originals, so net revenue is approximate; two large invoices (£77.2K and £168.5K) and their reversals distort monthly gross sales. Unknown customers account for 24.8% of lines, limiting retention analysis. Original SQL counted cancellation invoices as orders (23,796 vs 19,773), lowering AOV (£409.65 vs £518.24) and raising the repeat rate (70.0% vs 65.3%). Single-year coverage prevents year-over-year growth or confirmed seasonality.

10. Actionable Recommendations (ranked)

#	Evidence	Action	KPI / priority
1	1,449 customers inactive >90 days; £871,183 observed spend	Win-back campaign for top inactive spenders with a control group	Reactivation rate; High
2	Top 1% of customers = 30.2% of identified spend	Account management and monthly cadence monitoring for top customers	Top-customer order cadence; High
3	Cancellations 4.6% of gross sales; two large reversals	Capture reason codes and link cancellations to originals; review large orders	Cancellation rate by month; High
4	UK = 84.8% of revenue; non-UK markets small	Test growth in Germany/France (broadest bases) before high-AOV, low-customer markets	Non-UK share, customers per market; Medium
5	Sep-Nov drives much of the year; Dec 2011 partial	Plan stock and marketing for Q4; collect more years to confirm seasonality	Q4 orders vs capacity; Medium
6	Top 20% of codes = 78.4% of revenue; long tail	Review low and negative-net tail items for range decisions (revenue only, not profit)	Revenue per stock code; Medium
7	Max-price discount proxy implausible; no cost data	Source price list, promotion log and unit costs	Data availability; Medium

Success criteria are not quantified because the data provides no baseline targets; set them after agreeing baselines with the business.

11. Methodology and Metric Definitions

Classification: cancellations = InvoiceNo starting with C (all negative quantities); service codes listed above. Product lines drive all KPIs. Orders = distinct sales invoices; AOV = gross sales / orders; customer metrics use non-null CustomerID; observation date for inactivity = 2011-12-09. Growth = (current − prior)/prior with N/A for missing or zero prior. Reconciliation: product net £9,771,318.16 + service net −£23,187.09 = original SQL total £9,748,131.07 (difference 0.00). The calculations were run on the CSV with pandas; the SQL was not executed on SQL Server. Dashboard figures were checked against an independent pandas calculation (overall and for a Germany/2011 filter).

12. Final Management Conclusion

The business grew strongly into Q4 2011 on order volume, with a broad product base, but is exposed to one market and a small group of large customers, and its data cannot yet support profit, discount or seasonality claims. Next analytical steps: add cost and price-list data, link cancellations to originals, extend history for year-over-year comparison, and test win-back and cross-market initiatives.